Karel Capek
The Czech writer who first used the word "Robot" in the 1921 play RUR

Unimates
In the early '50s, George Devol and Joe Engleberger created the first modern industrial robot

Robotics
The word first appeared in 1942 in the short story "Runaround" by Isaac Asimov

years along with social / interactive, consumer and medical robots.

Mobile robots are utilised in industrial, military and security environments. There are several consumer robots meant for entertainment purposes, and some designed to perform certain domestic tasks such as floor cleaning. Autonomous robots with reasoning capabilities and the ability to move around freely will be in demand in the coming decades. Designing autonomous mobile robots in any meaningful degree has become possible only with the recent surge in computational, communications and sensing technologies. Teams of smart micro-robots could do regular maintenance in nuclear power plants and other hazardous environments.

Personal robots will change our lifestyle altogether by assisting us in our disabilities and nursing us in sickness. Artificial dogs or robot-pets with emotions can soothen many, especially children, and the aged and disabled.

While all this is exciting to experience, behind the scenes, there are several issues that crop up – where am I (positional information), where should I go (situation awareness and target identification), what should I do (target identification, object manipulation and reactive capabilities). These are issues that should be addressed.

When it comes to real-world operational conditions of mobile robots, the level of accuracy is also important. The most advanced species on this planet (humans) perform well out there with less precision and accuracy. Of course, it is desirable to have mobile robots that are capable of pin-point accuracy, which will depend on the application areas. Human capabilities around "tracking" and "following" are commendable, though precision and accuracy are not major concerns as we are comfortable in continuously adjusting our actions. There is a long way to go in bringing robots to the level of human-like capabilities.

Humanoids

We have built the environment that is suitable for two-legged systems. It is predicted that robots will be with us in our daily life sharing our space and resources (power, bandwidth and space). Nature has shown the way where the most successful species on this planet has two legs. And, the robots that may have to live with us in due course of time should preferably be two legged, or else, we'll have to redesign our living space to suit wheeled robots.

The most challenging issue with humanoids is balancing on two legs. Humans are capable of doing all kinds of acrobatics with two legs. We have muscles to assist us in coordinating various activities and the body is flexible. Research along material science (flexible body, muscles, actuators), nano technology (smaller and lighter sensors and actuators), computational intelligence (fuzzy logic, neural-networks, learning – genetic algorithms, evolutionary algorithms), should be assimilated into and mastered to design better systems.

GENUS

This 21 degrees of freedom humanoid robot is marketed by Robhatah Robotic Solutions, Bangalore. A single DSP-based controller board drives the servos. A camera, accelerometer and digital compass are connected to the DSP board for sensory feedback. The on-board accelerometer is used as a tilt sensor that allows the robot to detect a fall or dive and autonomously recover back into an upright standing posture. Trajectory-based gaits for the humanoid is generated in real time by reading predefined joins data and interpolate using cubic polynomial trajectory planning.

Robots (and humans) enjoying a game of soccer

The CREO

The Mechatronics and Automation laboratory of National University of Singapore developed this humanoid robot, and supports a wide range of research in robotics, including intelligent system, image processing, kinematics and dynamics, real-time control system, wireless communication and energy efficiency in biped locomotion. CREO stands tall at 540 mm with 26 degrees of freedom.

A double knee joint configuration based on closed kinematics is utilised. The actuators can be configured with different response profiles to fully maximise motion efficiency and effectiveness. The actuators use the daisy chain network topology. The humanoid sparks an open architectural platform in which the processors can be customised. It is armed with a 1.3-MP camera and an inertia measurement unit consisting of a three-axis accelerometer and two-axis gyro.

Traditionally, humanoid robots are often built with a lower centre of gravity (for better stability performance) and larger foot. CREO has a foot-body ratio closer to that of the humans and higher centre of gravity. A great deal of studies can be conducted in CREO on generating motion similar to that of a human in terms of dynamics, control and stability performance since the physical system is now closer to that of a human (ratio and weight distribution aspects).

Cooperative robotics

Cooperation / coordination among members was so much the need of the hour,